Cybersecurity Assignment 2: Network Scanning and Traffic Analysis

Total Marks: 20

Objective

To perform network discovery, port scanning, and traffic analysis on a local network using:

- Nmap
- Wireshark

Lab Setup

- Router (connected to network)
- Smartphone (connected to the router via Wi-Fi)
- PC/Laptop with Nmap and Wireshark installed

Task 1: Network Discovery (5 Marks)

Scan the network to identify connected devices.

Command:

```
nmap -sn 192.168.1.0/24
```

Requirements:

- Identify the **IP address of the phone**
- Identify the **router IP address**

Evidence:

- Screenshot showing discovered devices

Task 2: Port Scanning (5 Marks)

Scan the phone to identify open ports and services.

Command:

```
nmap -sV <PHONE_IP>
```

Requirements:

- List **at least 2 open ports**
- Identify **services running on those ports**

Evidence:

- Screenshot of scan results

Task 3: Packet Capture and Analysis (5 Marks)

Capture and analyze network traffic.

Steps:

1. Open Wireshark
2. Start capture on your active network interface
3. Use the phone to browse the internet
4. Stop capture after ~1–2 minutes

Requirements:

- Identify **ARP traffic**
- Identify **TCP traffic involving the phone IP**

Filter Examples:

```
arp
ip.addr == <PHONE_IP>
```

Evidence:

- Screenshot showing ARP packets
- Screenshot showing traffic from the phone

Task 4: Documentation and Analysis (5 Marks)

Write a short explanation (1–2 pages) covering:

- Devices discovered on the network
- Open ports and their risks
- Type of traffic observed in Wireshark
- What you learned from the lab

Submission Instructions

1. Combine all results into a **single PDF document**:
    - Screenshots
    - Explanations
2. Name the file:
3. `Assignment_2_Lab_Report.pdf`
4. Upload the PDF to a **GitHub repository** using GitHub